BDA Staff Graded Assignment 2

2024eb01487

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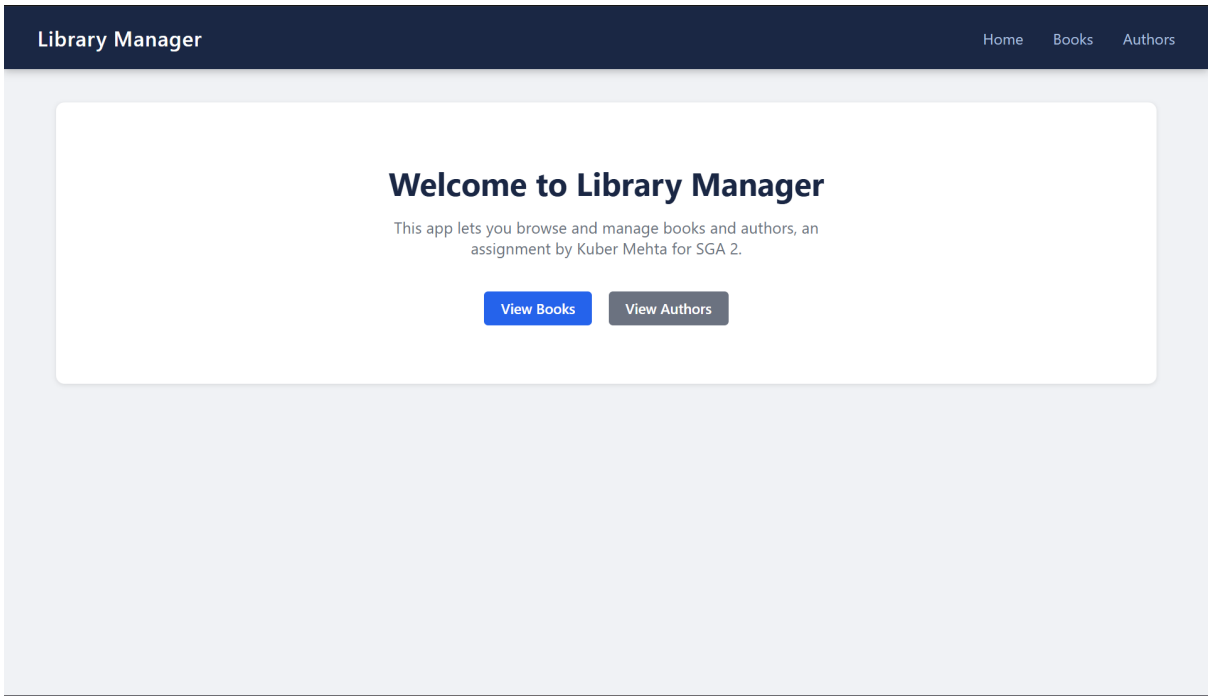
Library Manager App with Spring Boot



Approach

The application manages two entities: Author and Book. A book belongs to one author (Many-to-One), and an author can have many books (One-to-Many).

I picked some books and authors I liked and the ones that allowed me to represent a realistic, well-defined relationship.

The stack we're using is Spring Boot 2.7 with an H2 in-memory database, JSP views with JSTL, and JPA/Hibernate for ORM. The project is packaged as a WAR with embedded Tomcat so it can be run directly with `mvn spring-boot:run`.

E-R Design

Author (one) ----< Book (many)

Author table (authors):

- id (PK, auto-generated)
- name (NOT NULL)
- nationality
- birth_year

Book table (books):

- id (PK, auto-generated)
- title (NOT NULL)
- genre
- published_year
- isbn (UNIQUE)
- author_id (FK -> authors.id)

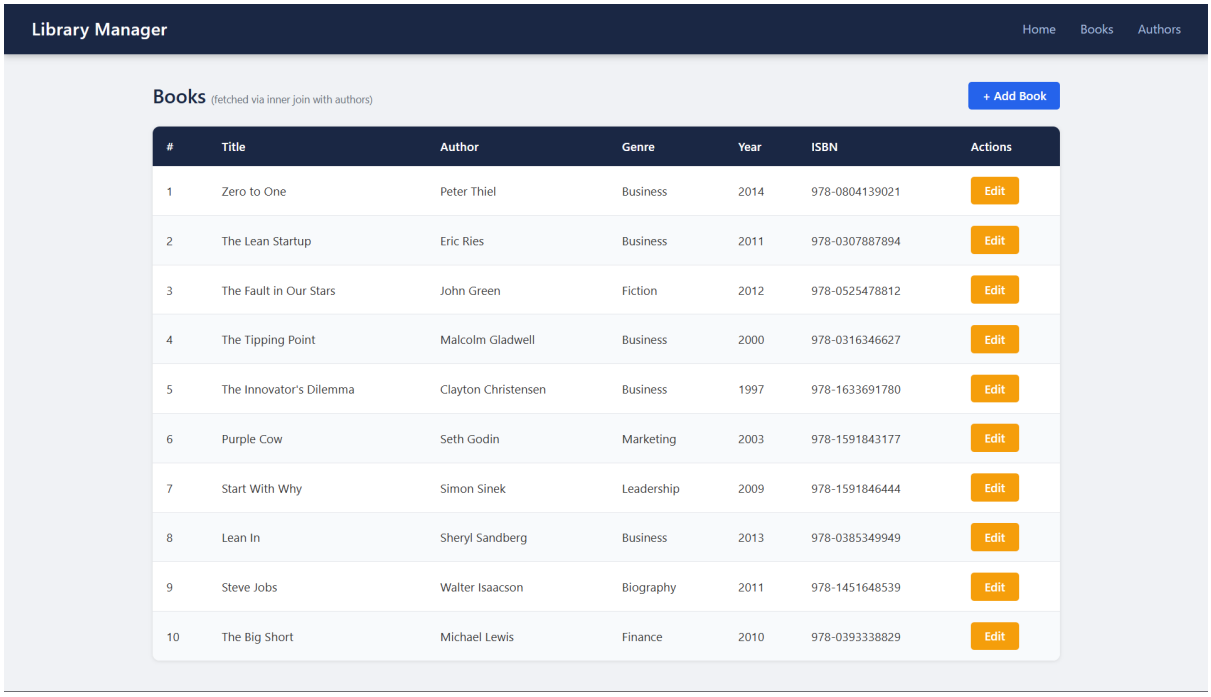
The OneToMany for the Authors and ManyToOne for the Books have a bidirectional relationship.

The unique constraint in this problem though is on isbn which is used to handle integrity violation.

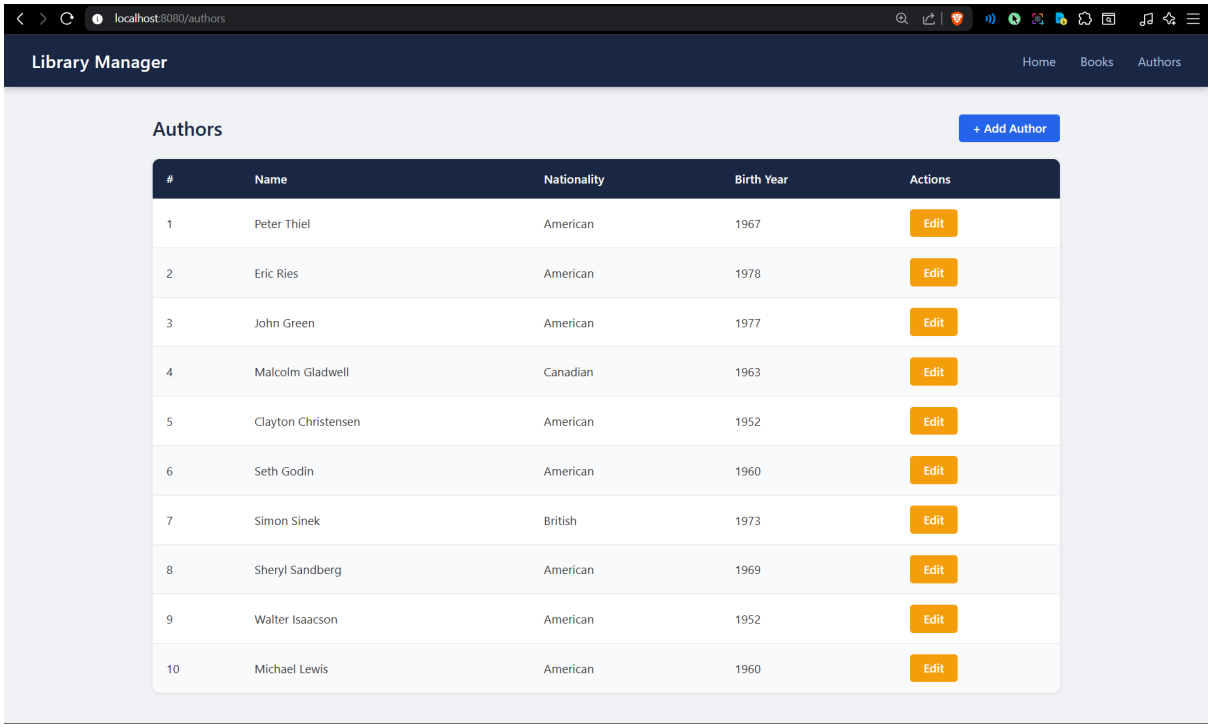
Implementation Details

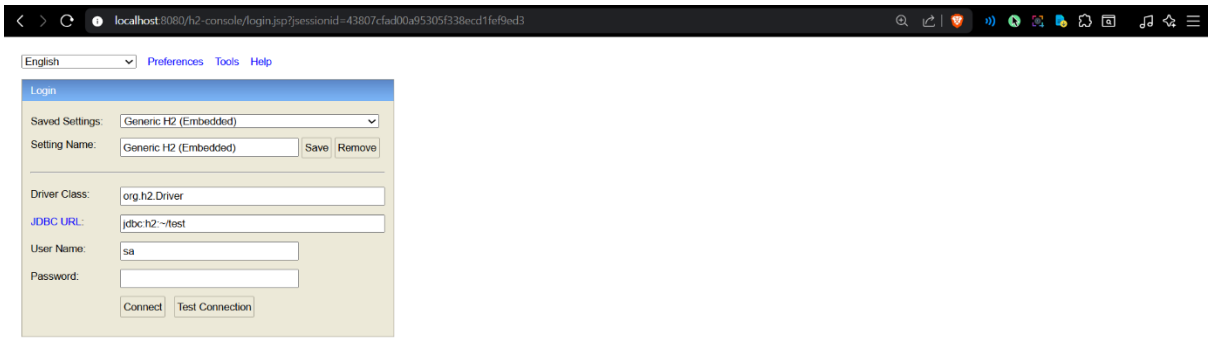
Populate Database

The DataInitializer class implements CommandLineRunner and runs on startup. Doing this we can insert 10 authors and 10 books using the repo layer.



#	Title	Author	Genre	Year	ISBN	Actions
1	Zero to One	Peter Thiel	Business	2014	978-0804139021	Edit
2	The Lean Startup	Eric Ries	Business	2011	978-0307887894	Edit
3	The Fault in Our Stars	John Green	Fiction	2012	978-0525478812	Edit
4	The Tipping Point	Malcolm Gladwell	Business	2000	978-0316346627	Edit
5	The Innovator's Dilemma	Clayton Christensen	Business	1997	978-1633691780	Edit
6	Purple Cow	Seth Godin	Marketing	2003	978-1591843177	Edit
7	Start With Why	Simon Sinek	Leadership	2009	978-1591846444	Edit
8	Lean In	Sheryl Sandberg	Business	2013	978-0385349949	Edit
9	Steve Jobs	Walter Isaacson	Biography	2011	978-1451648539	Edit
10	The Big Short	Michael Lewis	Finance	2010	978-0393338829	Edit


localhost:8080/authors
Library Manager
Home
Books
Authors
Authors
+ Add Author
# Name Nationality Birth Year Actions
1 Peter Thiel American 1967 Edit
2 Eric Ries American 1978 Edit
3 John Green American 1977 Edit
4 Malcolm Gladwell Canadian 1963 Edit
5 Clayton Christensen American 1952 Edit
6 Seth Godin American 1960 Edit
7 Simon Sinek British 1973 Edit
8 Sheryl Sandberg American 1969 Edit
9 Walter Isaacson American 1952 Edit
10 Michael Lewis American 1960 Edit

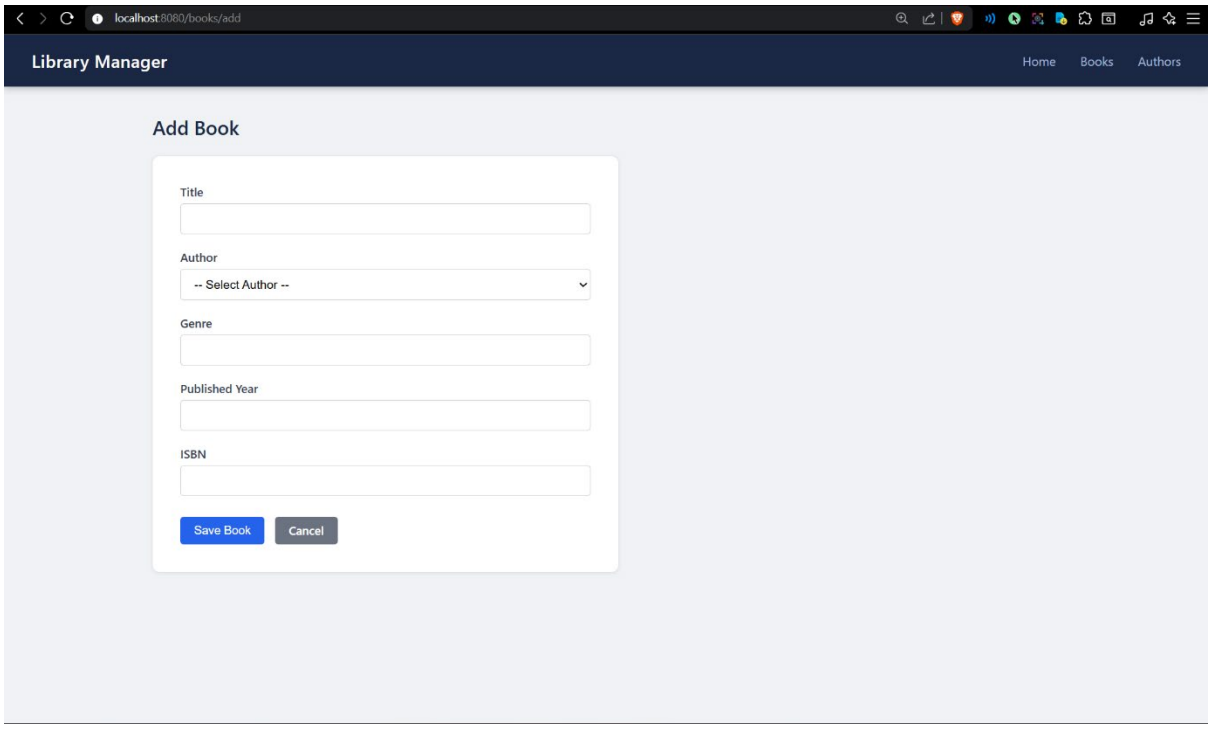

localhost:8080/h2-console/login.jsp?jsessionid=43807cfad00a95305f338ecd1fef9ed3
English
Preferences
Tools
Help
Login
Saved Settings:
Generic H2 (Embedded)
Setting Name:
Generic H2 (Embedded)
Save
Remove
Driver Class:
org.h2.Driver
JDBC URL:
jdbc:h2:~/test
User Name:
sa
Password:
Connect
Test Connection

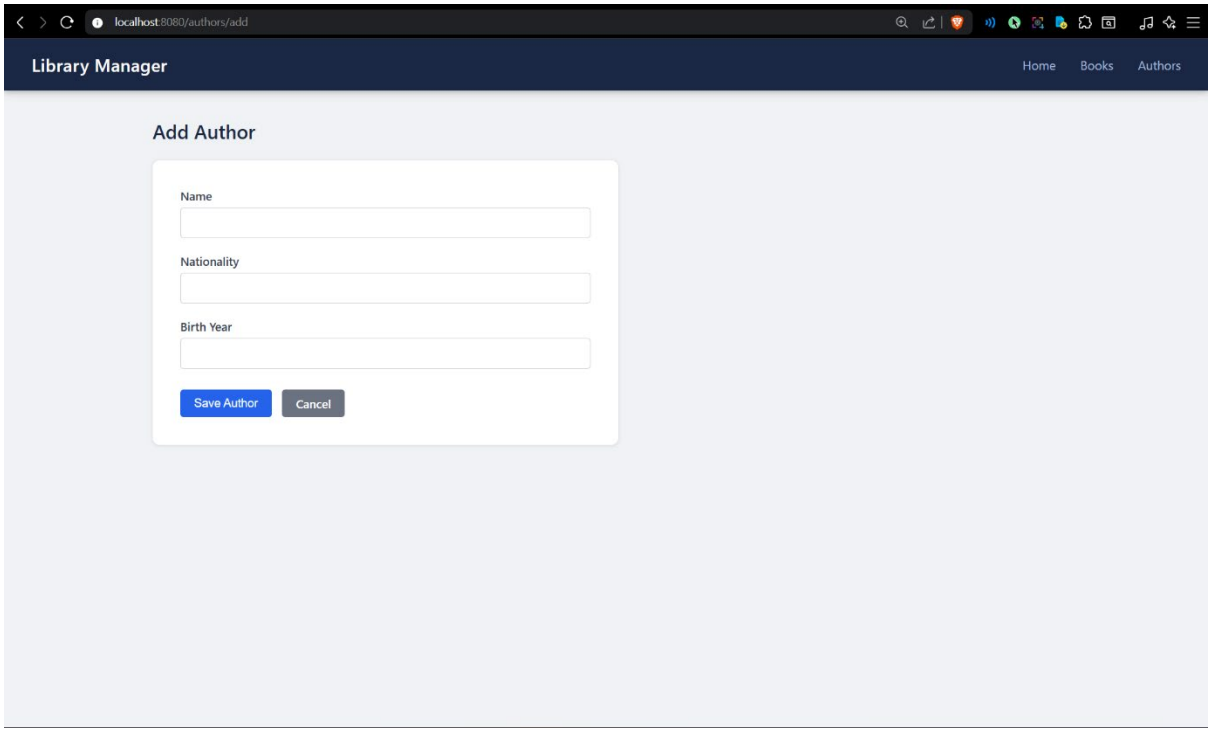
Create Operation

Controller method in BookController handles POST commands in /books/add. It receives the form fields via @ModelAttribute and the selected author ID via @RequestParam, fetches the Author, sets it on the book.

If a DataIntegrityViolationException error is thrown, the error message is added to the model and the form is re-rendered.

The same idea applies to AuthorController for POST /authors/add.





Read Operation

The books list page calls bookService.getAllBooksWithAuthors(), which calls the custom JPQL query in BookRepository:

```
@Query("SELECT b FROM Book b INNER JOIN b.author a")
List<Book> findAllBooksWithAuthors();
```

INNER JOIN makes sure that only books that have an associated author are returned.

The result is bound to the JSP view and rendered in a table showing title, author name, genre, year, and ISBN.

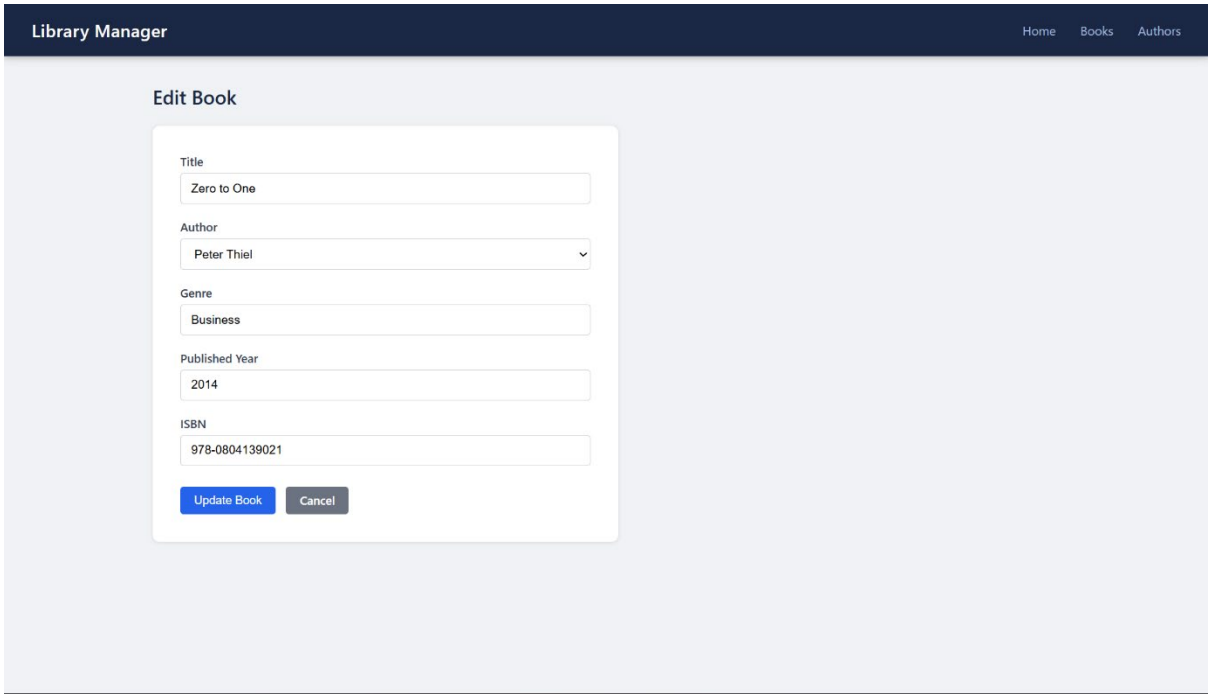
Update Operation

Clicking Edit on any row redirects the user to /books/edit/{id} or /authors/edit/{id}.

The controller fetches the existing entity, adds it to the model, and renders the edit form with fields pre-filled. For the book edit form, the author dropdown pre-selects the current author using a JSTL conditional:

```
<option value="${a.id}" <c:if test="${book.author != null && book.author.id == a.id}">selected</c:if>>
```

On POST, the controller makes it so the ID is on the model object before saving, so Hibernate performs an UPDATE instead of INSERT.



Testing

We wrote comprehensive tests for this project too, the repo tests use @DataJpaTest which have a slice context with H2. They test save, findById, the custom inner join query, and the behavior that books without an author are excluded from the INNER JOIN result.

Service tests use @ExtendWith(MockitoExtension.class) and uses mockito. They test that the service delegates correctly and verify interactions.

The entire set of tests can be run with just: mvn test

```
PS X:\Assignments\BDA-SGA2> mvn test
Customizer@2cb4893b, org.springframework.boot.test.mock.mockito.MockitoContextCustomizer@0, org.springframework.boot.test.autoconfigure.OverrideAutoConfiguratio
nContextCustomizerFactory$DisableAutoConfigurationContextCustomizer@6e9a5ed8, org.springframework.boot.test.autoconfigure.actuate.metrics.MetricsExportContextCu
stomizerFactory$DisableMetricExportContextCustomizer@31ea9581, org.springframework.boot.test.autoconfigure.filter.TypeExcludeFiltersContextCustomizer@351584c0,
org.springframework.boot.test.autoconfigure.properties.PropertyMappingContextCustomizer@b7bce998, org.springframework.boot.test.autoconfigure.web.servlet.WebDri
verContextCustomizerFactory$Customizer@4bdeaabb, org.springframework.boot.test.context.SpringBootTestArgs@1, org.springframework.boot.test.context.SpringBootTes
tWebEnvironment@0], contextLoader = 'org.springframework.boot.test.context.SpringBootContextLoader', parent = [null]], attributes = map['org.springframework.tes
t.context.event.ApplicationEventsTestExecutionListener.recordApplicationEvents' -> false]]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.051 s - in com.bda.sga2.repository.BookRepositoryTest
[INFO] Running com.bda.sga2.service.AuthorServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.105 s - in com.bda.sga2.service.AuthorServiceTest
[INFO] Running com.bda.sga2.service.BookServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.023 s - in com.bda.sga2.service.BookServiceTest
2026-05-02 17:41:10.283  INFO 31420 --- [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'defa
ult'
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 16, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  7.772 s
[INFO] Finished at: 2026-05-02T17:41:10+05:30
[INFO] ------------------------------------------------------------------------
PS X:\Assignments\BDA-SGA2>
```

Challenges

1. Author dropdown in the book form - @ModelAttribute only handles flat form fields, so it chokes on nested entities like Author. The fix was to pull the author ID in as a plain @RequestParam Long authorId and fetch the entity manually in the controller.

2. Pre-selecting the author in the edit form - used a JSTL c:if tag to check each dropdown option against the current book's author ID as it loops through.

GitHub Repository URL

https://github.com/Kuberwastaken/bda-sga2